CAP6412
Assessing Ambiguity in Vision-Language Instructions

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Problem Description

- Models do not understand Language the way that humans understand it
- Ambiguity in Language compounds the problem
- Some instructions from humans are not clear enough to be performed

Approach / Plan

- BLIP-2 backbone
  - Can produce strong multimodal embeddings
  - Input: Scene image, user instruction
  - Output: fused multimodal embedding
- Classify ambiguity
  - On top of BLIP; MLP or small transformer head
    - Possible extensions
  - Prompt: “Move that away”; Model Output: “Ambiguous” or “Not Ambiguous”
- Clarification Question
  - If ambiguity detected:
    - Template-based using scene

Dataset

Language instruction, visual scene information:

- TEACh
  - Human-human interactive dialogues
  - Designed around interaction
- ALFRED
  - Natural language instructions
  - Visual observations
  - Sequences of actions
- DialFRED
  - Extension of ALFRED
  - Human-annotated task-relevant questions/answers

Metric

- Ambiguity accuracy / F1
  - Ambiguity type classification
- Clarification question relevance
- No clarification baseline
- Improvement after clarification

Contributions

- Shoshanah – Metric Evaluation
- Erick – Dataset Prep
- Rafael – Model Training
- Obinsonne – Model Training
- Sky - Research